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Lab Study 2

1) Ferromagnetic core with one air gap

$L_{core} = 0,18\,m$ mean mag. path of cor

$g = 0,50\,mm = 5 \times 10^{-4}\,m$ single air gap

$N = 400$ Turns

$\mu_0 = 4\pi \times 10^{-7}\ H/m = 1,256637 \times 10^{-6}\ H/m$ → vacuum permeability

$$B = \frac{\mu_0 . N . I}{\left(g + L_{core}/\mu_r\right)} \qquad B = S . I$$

$$S = \frac{\mu_0 . N}{\left(g + L_{core}/\mu_r\right)} \Rightarrow \mu_r = \frac{L_{core}}{\left(\mu_0 . N/S - g\right)}$$

| I (A) | B (T) |
|---|---|
| 0,10 | 0,085 |
| 0,2 | 0,170 |
| 0,4 | 0,340 |
| 0,6 | 0,510 |
| 0,8 | 0,680 |
| 1 | 0,850 |

These points are very linear, so the best fit slope is

$S \approx 0,85\ T/A$

Compute $\mu_r$ from the slop

$\mu_0 . N = 1,256637 \times 10^{-6} \times 400 = 5,0265 \times 10^{-4}$

$\mu_0 . N / S = (5,0265 \times 10^{-4}) / 0,85 = 5,9136 \times 10^{-4}$

Denominator terms $(\mu_0 . N/S) - g = 5,9136 \times 10^{-4} - 5 \times 10^{-4} = 9,136 \times 10^{-5}$

Relative permeability $\mu_r = 0,18 / 9,136 \times 10^{-5} = 1,97 \times 10^{3}$

$\mu_r = 1970$

Comments

If datasheets says $\mu_r$ theory $\approx 2000$, our estimat ($\approx 1970$) is within ~1-2%. Differences can come from:

- fringing higher B
- gap tolerans
- Leakage flux and non-unifor cross-section
- B(H) nonlinearith
- measurement uncer

Q2) Two gaps of different lenghts - flux split via reluctance

two parallel legs with gaps

Same core $\mu_r = 2000$

$A = 1.0 \times 10^{-4}\ m^2$

Leg 1 gap: $g_1 = 0,5\ mm = 5 \times 10^{-4} m$

Leg 2 gap: $g_2 = 1,5\ mm = 1,5 \times 10^{-3} m$

$L_{core,\ leg} = 0,16 m$

Excitation: $N = 400$, $I = 0,6\ A \rightarrow \mathcal{F} = NI = 240\ A.turns$

Reluctance of branch: $\mathcal{R} = L_{core,\ leg} / (\mu_0 \mu_r A) + (g / \mu_0 A)$

(core part per leg)

$\mathcal{R}_{core} = 0,16 / (\mu_0 \mu_r) = 0,16 / (1,256637 \times 10^{-6} \times 2000 \times 1 \times 10^{-4}) = 6,37 \times 10^5\ A/$

(Gap parts)

$\mathcal{R}_{g1} = 5 \times 10^{-4} / (\mu_0 A) = 5 \times 10^{-4} / 1,256637 \times 10^{-10} = 3,98 \times 10^6$

$\mathcal{R}_{g2} = 1,5 \times 10^{-3} / (\mu_0 A) = 1,5 \times 10^{-3} / 1,256637 \times 10^{-10} = 1,19 \times 10^7$

Flux:

$\Phi_1 = \mathcal{F} / \mathcal{R}_1 = 240 / 4,64 \times 10^6 = 5,17 \times 10^{-5}\ Wb$

$\Phi_2 = \mathcal{F} / \mathcal{R}_2 = 240 / 1,26 \times 10^7 = 1,91 \times 10^{-5}\ Wb$

Flux densities

$B_1 = \Phi_1 / A = 0,517\ T$ $\quad B_1 > B_2$ $\quad \frac{g_1}{g_2} = 3$

$B_2 = \Phi_2 / A = 0,191\ T$ $\quad \frac{B_1}{B_2} \approx 2,71$

small differ comes from shared core re... tru...

Two gaps, shorter gap carried ~2.7x more flux density; flux split inversely to branch reluctances. The equivalent reluctance and flux addition followed the same mathematics as resistors and currents in electric circuits.

CamScanner ile tarandı